

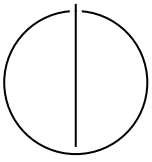
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
CHAIR OF HEALTH INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification

Lukas Alex Jochim





SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — CHAIR OF HEALTH INFORMATICS

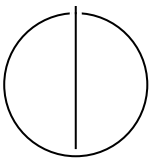
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Decoding Music Perception: EEG-Based Genre Classification

Dekodierung der Musikwahrnehmung: EEG-basierte Genre-Klassifizierung

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Submission Date:	01.09.2025



I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 01.09.2025 Lukas Alex Jochim

Acknowledgments

Abstract

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1 Introduction

1.1 Motivation

The human brain is one of the most complex and important systems within the human body. The brain manages various functionalities like emotion, memory, attention and sensory processing, to name a few. Apart from that, the brain plays a significant role in how we process musical pieces. Independent of culture and age, music is a universal tool to influence one's emotions, concentration and cognition. Furthermore, the brain gives us the unique ability to retrieve many different features out of music, like melody extraction or the identification of an artist or a specific genre. Depending on the style of music, the brain reacts differently. However, although the brain is central to cognitive processes such as listening to music, the core mechanisms remain unknown and are currently black boxes.

To enlighten these black boxes, medical advancements like functional Magnetic resonsance imaging (fMRI) or Electroencephalogram (EEG) arrived, which can measure brain activity. Especially EEG has aroused high interest since they directly measure the neural activity and can be done non-invasive with a little overhead in setup. This is why combining EEG and music has the ability to help better understand how our brain functions. By further adding means of Machine Learning (ML), a whole new interdisciplinary field opens up, bridging the gap between neurosciences and Computer science.

There is a growing research interest in the affective influence of music on the brain. However, classifying different music genres based on EEG signals remains underrepresented. This is exactly where this thesis aims to close the gap, by investigating whether it is possible to classify various musical genres depending on the EEG signals acquired using state-of-the-art ML techniques.

The research results have high potential in the domain of medical computer science. Especially with the growing investigations on Brain Machine Interfaces (BMIs), adaptive music therapy, personalised music recommendations or neurofeedback systems may become possible. Additionally, entertainment technologies can become more user-centred, facilitating unique experiences that impact many sectors like education and mental health

1.2 Research goals

1.3 Methodology

2 Theoretical Background

2.1 Medical foundations of EEG signals

2.2 Brain rythms

2.3 Correlation between music and brain activity

2.4 Recording and measurement of EEG signals

2.5 Denoising and preprocessing of EEG signals

2.6 Feature extraction and signal decomposition of EEG signals

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2.7.2 Validation and evaluation

3 Related work

3.1 Music Genre classification with audio features

3.2 EEG datasets with music stimulus

3.2.1 DEAP dataset

3.2.2 OpenMIIR initiative

3.2.3 MUSIN-G

3.3 Emotion classification with EEG signals

3.4 Music Genre classification with EEG signals

4 Music Datasets

Since this project deals with music genre classification, a suitable set of musical stimuli have to be chosen to gather valuable data. As research in the domain of music is widespread, there exist multiple common music datasets used in different studies. Therefore, the difficulty lies in choosing the right dataset for this thesis. However, the project goal cristallizes out clear requirements for such a music dataset.

First of all, the songs that will be listened to are suppossed to be around 30s long. The reason for that is that the stimulus could go lost if the song runs too long because the brain can adapt to the genre-specific features, thus reducing the visibility of the impact on the brain. Apart from that, the amount of songs should be quite large with around 800 to 1000 song clips. It is important to have a large pool of songs to get as many different and versatile data as possible. This helps in future steps regarding classification. However, related to this requirement, the genres used in this pool also have to meet specifications. First of all, the songs must be clearly labeled to avoid confusion in later stages of the study. This is also important to know for sure how balanced the dataset is in terms of genres listened to. This is supposed to be as balanced as possible so that each genre is represented equally. Additionally, it should be easy to employ this dataset in the study. It should be uncomplicated to load the songs as well as getting access to labels or other important information.

In the following, different popular and public music datasets are presented. They are going to be analyzed according their possible use in this research and on how they meet the requirements specified above. After that, the reasons for the final choice of dataset are outlined.

4.1 Audio Set

In 2017, Gemmeke et al. published a dataset for audio events called Audio Set[1]. This is a large-scale dataset consisting of audio events labeld by humans using a carefully structured hierarchical ontology of 632 audio classes. The source of the audio events are derived from 10 second segments of YouTube videos. These audio events are occurences in the real-world, like human sounds, animal sounds, natural sounds, sounds of things,

[1] `https://research.google.com/audioset/`

source-ambigious sounds, environment sounds and music. Then, each of these sounds is labeled with one ontology and labels with a maximal hierarchical depth of six. In the determination of labels, the researchers made sure that the tags are intuitive and clearly distinct from each other. [Gem+17]

In terms of the use for this project, the audio set has some drawbacks. First of all, the focus of this dataset does not lie in music but in general audio events. This is why in the first step, the dataset would have to be filtered to just contain the music clips. However, the fact that the clips are only 10 seconds long, there is the risk that this is not enough of a stimulus to arouse the brain and see the distinct patterns. Another problem lies in the format of the data. Since every audio event is a YouTube link, each sound would have to be downloaded from that link to allow a seamless integration with the experiment setup, thus illustrating a large overhead. Last but not least, the dataset is not labeled in regards of specific genres.WRONG CHANGE THIS: SEE THIS PAPER FOR REFERENCE: `https://arxiv.org/pdf/1804.01149` ; THEY HAVE USED IT FOR GENRE CLASSIFICATION Therefore this not only adds the overhead of manually mach each music piece to its genre but also does not guarantee that each genres appears the same amount of times in the dataset, making it a not a so well fit for the grounds of this thesis.

4.2 Million Song Dataset

The Million Song Dataset[2] is a collection of audio features and metadata for a million music tracks, which also is freely available. At the moment, this is the largest research dataset in the field as it contains 1,000,000 songs accumulating around 280GB of data. In addition to the songs mainly from western commercial music in 2011, general metadata and audio features like pitch, timbre, etc. are provided by making use of the EchoNest API. This API is an online resource supporting many music applications on the web. Since most of the "natively" available metadata are coming from the EchoNest API, it is easily extendable to further labels like genres. [Ber+11]

Although this dataset may seem like a great fit because of its rich amount of songs, the amount of songs is too much. Since only 800-1000 songs are used during the acquisition, a majority of the dataset would have to be filtered out. Furthermore, all songs are not trimmed down to a shorter clip but are provided in their full length, which is not meeting the requirement for this study. Furthermore, the genres would have to be added externally since they are not available in the native dataset. This also emphasizes that it does not guarantee any balance in genres available over the course of the 1,000,000 songs.

[2] `http://millionsongdataset.com/`

4.3 MagnaTagATune

One challenge that comes up when creating such datasets is how a label or tag is determined for each song. In the context of genres, how is it decided which genre which song belongs to, as there could be overlapping features. On the basis of this hurdle, Law et al. created a music dataset to evaluate different tagging algorithms called MagnaTagATune[3]. It contains 25,863 music clips each 29 seconds long. These excerpts were taken from 5223 songs. These music pieces range across many different genres like Rock, Pop, Jazz, Blues, World, Metal, Punk and many more. In general, each clip is accompanied by a vector of binary annotations of 188 tags using the TagATune game. [Güç+16]

TagATune is a game with two players that aims to collect music tags from the players. Each player is presented with a music clip which they are assigned to give them labels. After that, both players see what the other player has chosen as tags. Based on this information, the players have to decide whether they listened to the same music clip. [Law+09]

In perspective of this study, this music dataset has the advantage that each clip is around 30 seconds long, hitting the desired length for one stimulus. Furthermore, it is beneficial that the songs vary in genres used, thus being a good candidate as a music dataset. However, there are some drawbacks, especially lying in the labeling of each clip. Although the labeling process is well-thought and transparent, each song can have more than one label. This means that, depending on the players, one clip can be annotated with more than one genre tag. And since there are in total 37 genre related tags [Rot23], deciding which genre is the most suitable choice for a song is very difficult. Related to that, since there are around 26,000 clips, one would have to remove majority of songs in order to let it fit into the scope of this thesis. Combined with the fact that multiple genres can be chosen for each song, it is difficult to achieve a balanced dataset according in regards of genre.

4.4 FMA Dataset

The Free Music Archive (FMA)[4] is an open dataset designed for multiple tasks in the domain of Music Information Retrieval (MIR)

[3] `https://paperswithcode.com/dataset/magnatagatune`
[4] `https://github.com/mdeff/fma`

4.5 GTZAN Genre classification

4.6 Final decision

Add a table summarizing the datasets. As columns, it should have name, amount of songs/music clips, average length of songs, amount of genres, availability of audio files

5 Methodology

5.1 Acquisition

5.1.1 General setup, environment and stimuli

The experiment has been conducted in a laboratory setting. As mentioned in Chapter 4, the GTZAN Music dataset is chosen as a basis for the music stimuli. However, for the sake of the quality of the data, it would not be beneficial to hear all 1000 songs in one session. To avoid quick attention loss during a session and gather as much different data under different circumstances, the whole set of audio files is partitioned into 20 distinct sessions, each lasting around 30 minutes. This means that per session, 50 songs are played[1]. For each of these songs, not only are EEG signals measured, but also other biosignals like Photoplethysmogram (PPG), Inertial Measurement Unit (IMU) and temperature. However, this work will solely focus on the recorded EEG data. However, these 50 songs are not chosen randomly. Five songs from each genre are being played during the session to ensure that each of the ten genres from the dataset is represented in each session. This also ensures no genre bias in the data, thus resulting in a balanced signal across genres. During listening, the order of the songs is randomised to evoke a brain reaction that is as natural as possible, as the brain cannot predict the genre that is being listened to. Additionally, a five-second break is implemented in between songs to not overstimulate the brain.

All 20 sessions have been recorded over two weeks, with an average of two sessions per day. Between each session, there was a short break of between ten and 30 minutes. This verifies a sufficient pause of the brain to fully prepare for the next session. The test person is the author, a 21-year-old, right-handed college student without any common (brain) diseases. To further guarantee a high quality in the acquired signal data, the experiment's environment and setting must be adapted. First and foremost, the test person is instructed to sit comfortably in a chair with his eyes closed to minimise artificats coming from blinking and eye-movement. Furthermore and most importantly, any movements, especially in the head region, must be limited and as minimal as possible since the EEG device reacts very sensitive to head movements. Apart from that, the room was quiet and isolated with dimmed natural lighting combined with the

[1] Because of a corrupt audio file, one session only consists of 49 songs

brightness from a desktop monitor. The songs were played over the EEG headset (cf. Subsection 5.1.2) with average and constant volume.

5.1.2 Acquisition device

A non-traditional approach is chosen to record the EEG signal data coming from the brain. For this study, a prototype for a hybrid biosensing platform *Hugin* by Jabra is chosen. It is a Bluetooth headset that can play music and measure biosignals at the same time. For Audio, it uses classic Bluetooth to connect to a standard PC or Laptop. For the data, however, it uses a second low-energy bluetooth channel for the biosignal data, which can be connected to a second data acquisition PC.

Add an image of the headset The headphones have multiple sensors and electrodes to measure biosignals like EEG, PPG, IMU and temperature. For EEG, the headphones are supplied with 2 × 4 sensor electrodes and two bias electrodes. The sensors are easily configurable and provide an ultra-low-noise front-end, meaning that the data points are amplified with minimal distortion and interference. Besides that, the electrodes sample with 24-bit and a frequency of 250Hz.

For PPG, one optical sensor on the right side provides a high resolution of 19-bit ADC with a configurable sample rate. They also support raw data output or built-in algorithms to conclude heart rate or SpO2.

The IMU data is sensed by a 3-axis digital accelerometer fused with a 3-axis digital gyroscope. On a full scale, the accelerometer can recognize acceleration from $\pm 2g$ up to $\pm 16g$ (in steps of the power of two). The gyroscope can measure angular velocity from $\pm 125°/s$ up to $\pm 4000°/s$. The recognised data is then combined to provide accurate information on orientation, movements and position.

Last but not least, there are three digital thermometers, one on the skin, another one inside the earcup and one on the outside of the earcup to measure ambient temperature. All of those temperature sensors have an accuracy of ± 0.1°C.

Since this headset is still a prototype, they still is ongoing development and research on the headset. Nonetheless, some small-scale research projects already employed the biosignals from the headphones. These projects range from stress detection using the PPG data to device control and the recognition of cognitive fatigue using the EEG sensors.

5.1.3 Technical setup

Multiple technical components have to come together to gather the experiment's data. All the components employ the Lab Streaming Layer (LSL). It provides a standardised

way to stream time-sensitive data between devices and applications. Devices or apps then push data on this stream, which is captured and stored in a special file format.

At the core, there are two main programs responsible for the data. On the one hand, there is the Headset platform called *BioGNee*. This program is responsible for the sensor setup and starting and stopping the signal acquisition. Communicating with the headset's low-energy Bluetooth Channel reliably sets up the biosensors of the hybrid headphones, making the device ready for acquisition. After a successful setup, the data is pushed on the corresponding LSL Stream. On the other hand, the study uses a custom Python script. This Python script has two main tasks. Firstly, it loads all the songs from one of the 20 pre-generated sessions and randomly shuffles them. After that, the program plays each song one by one. Simultaneously, the script also pushes timestamps during recording to assign each segment to each song later.

The LSL Recorder combines both of these components. The LSL recorder collects all open streams on which the device/programs push their data. These are collected and stored in an XML-like file called .xdf. The resulting document, therefore, provides a unified way to not only store the signal data but also the respective time markers. These can be visualised by a custom MATLAB script to inspect the data quality. Important to note is that each distinct session occupies one .xdf file.

Add a picture that visualizes the flow of acquisition and the communication between each part of the system For future steps like Denoising or Feature Extraction, it is essential to look at each segment individually, i.e., each song. Therefore, a further custom Python program is used for the segmentation process of one session. Since the start and the end of one song are pushed to the LSL stream, one can read out the time these markers have been pushed and fetch all the EEG data points measured in this interval. These segments are then written in a new .xdf file to allow easy handling in future processing steps and seamlessly work with the existing MATLAB script for EEG visualisation. CHANGE THIS SINCE THE IMPLEMENTATION CHANGED. HAS TO BE UPDATED

5.2 Denoising

5.3 Feature extraction

5.4 Classification

5.4.1 Model 1

5.4.2 Model 2

5.4.3 Model 3

6 Results

6.1 Metric 1

6.2 Metric 2

6.3 Metric 3

7 Conclusion

7.1 Key findings and results

7.2 Problems and Limitations

7.3 Future work

Abbreviations

fMRI functional Magnetic resonsance imaging

ML Machine Learning

BMI Brain Machine Interface

FMA Free Music Archive

MIR Music Information Retrieval

EEG Electroencephalogram

PPG Photoplethysmogram

IMU Inertial Measurement Unit

LSL Lab Streaming Layer

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